Debt Securities: Bonds

Chapter 5

Introduction to Financial Management (Pope)

What is a Bond (Debt)?

A bond is a security that obligates the issuer to make specified payments to the bondholder

- provide periodic interest income – annuity series
- return of the principal amount at maturity – future lump sum
- prices can be calculated by using present value techniques (discounting of future cash flows)
- first claimant

Indenture Agreement

Merrill Lynch As of 15-Jul-2008

OVERVIEW

Price (% of par):	109.13
Coupon rate:	6.50%
Maturity date:	15-Jul-2018
Yield to maturity:	5.30%
Current yield:	5.96%
Fitch ratings:	AA
Coupon payment frequency:	Annual
First coupon date:	15-Jul-2009
Type:	Corporate
Callable:	No

Par value (Face Value): payment at maturity of bond, typically $1000

Coupon rate: annual rate of interest paid as a percentage of par value

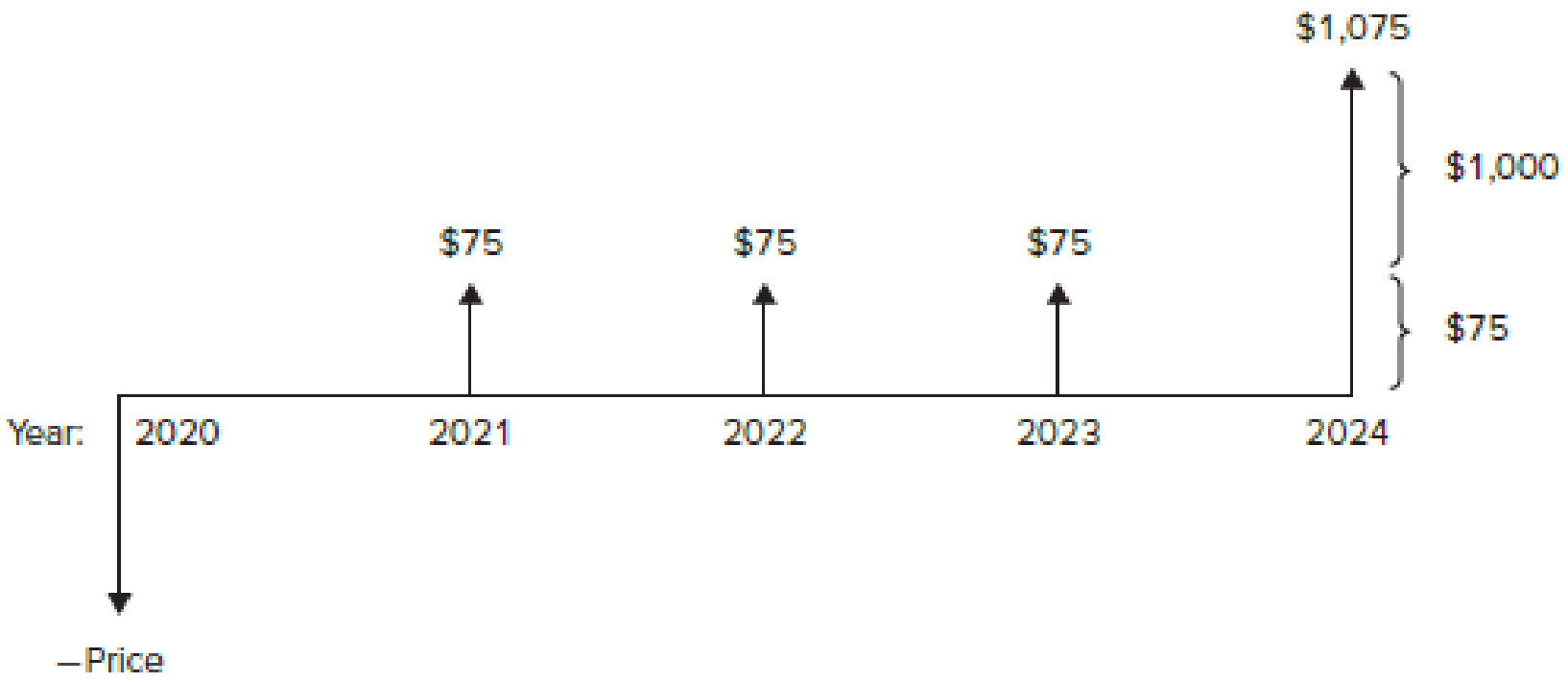
Coupon: regular interest payment received by holder (the annuity)

Maturity date: expiration date of bond when par value is paid back.

Yield to maturity: expected rate of return based on price of bond

Example 1:

Below are the cash flows to an investor for a 4 year, 7.50% coupon annual bond maturing in 2024 with a par value of $1,000.



Bond Pricing

$$PV = C\left(\frac{1}{r} - \frac{1}{r(1+r)^T}\right) + \frac{FV}{(1+r)^T}$$

- Discount the value of the coupon payments (the annuity)
- Discount the value of the face value (the lump sum)
- Add them together
- C vs. r

Example 2:

Calculate the price of an AAA-rated, 20-year, 8% coupon (paid annually) corporate bond (par value = $1,000) which is expected to earn a yield to maturity of 10%.

Semi-annual and Zero Coupon

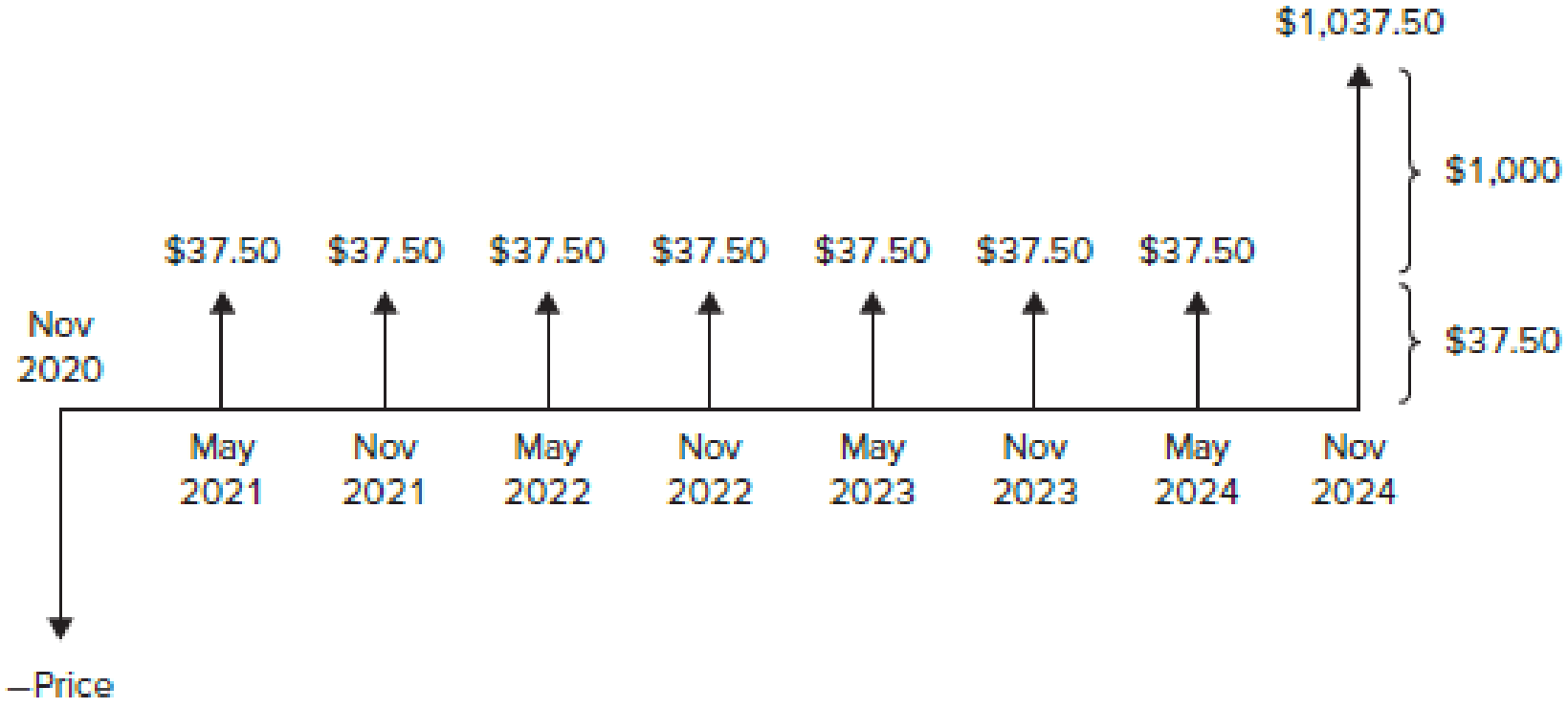
Most corporate and government bonds pay coupons on a semiannual basis (twice per year)

Some companies issue zero-coupon bonds by selling them at a deep discount

For computing price of these bonds, the values of the inputs have to be adjusted according to the frequency of the coupons (or absence thereof).

Example 3:

Here are the cash flows to an investor for a 4 year, 7.50% semi-annual coupon bond maturing in 2024. The bond pays semi-annual coupons, $37.50 twice each year and had a yield to maturity of 3%. What is its current price?



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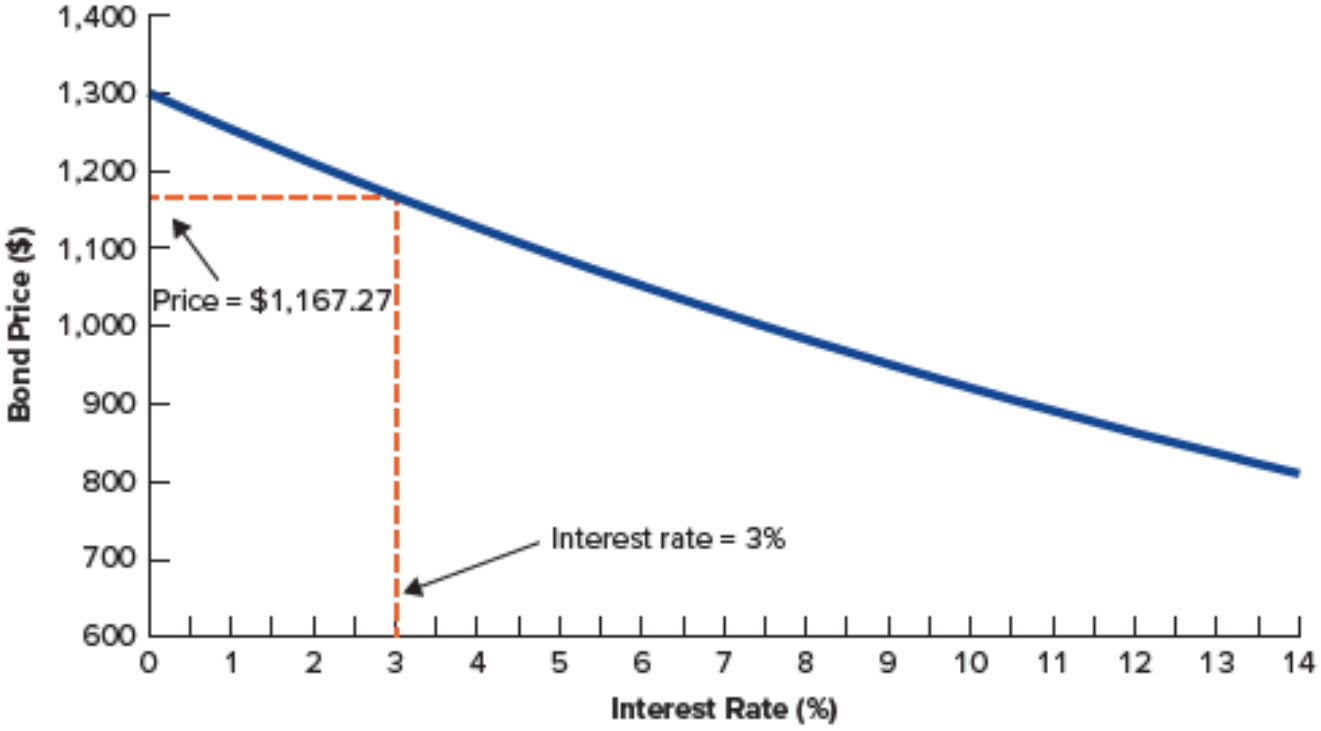
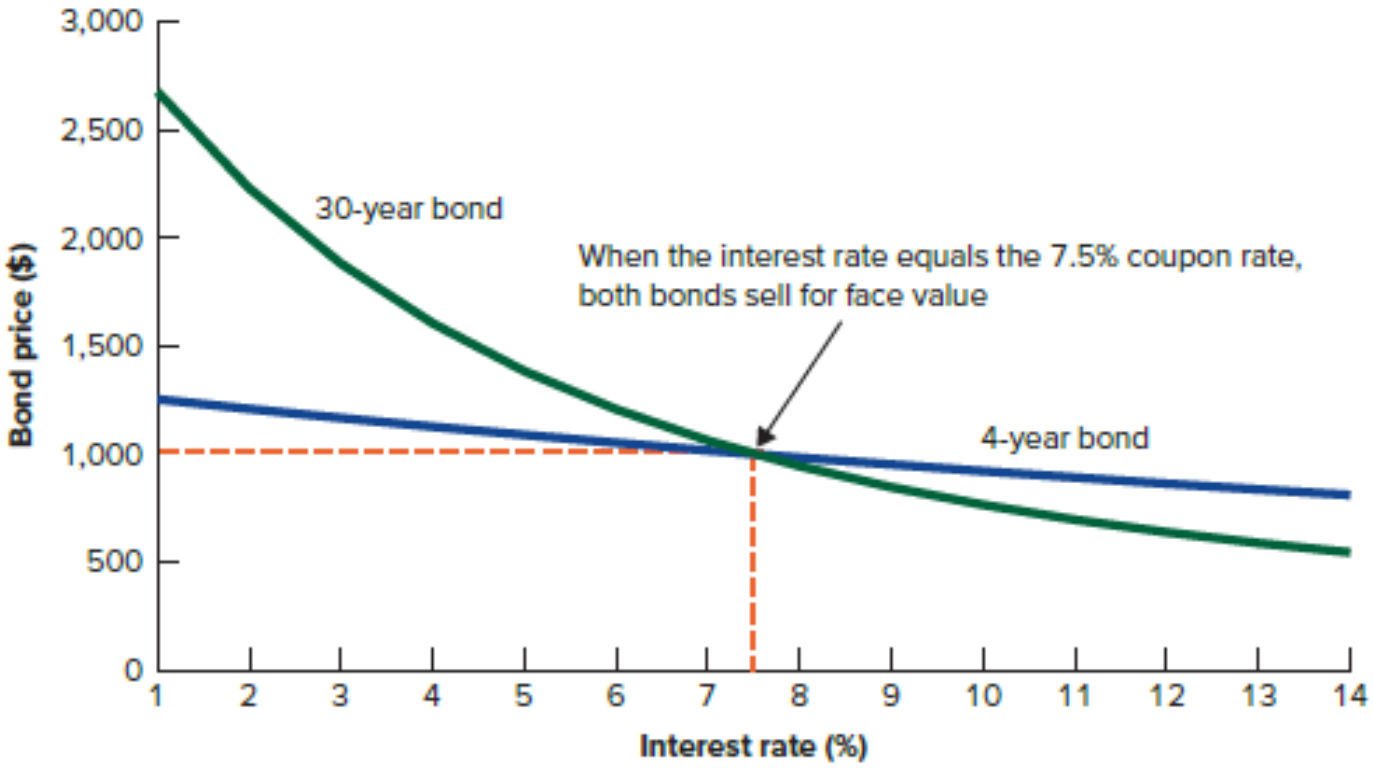


Interest Rates and Bond Prices

Interest rate risk: A change in interest rates impacts long-term bonds more than short-term bonds.



Yield to Maturity

Discount rate for which the present value of the bond's payments equals the price.

- **Expected rate of return on a bond if held to maturity.**
- Changes in economic conditions and risk factors will cause bond prices and their corresponding YTMs to change

$$PV = C\left(\frac{1}{r} - \frac{1}{r(1+r)^T}\right) + \frac{FV}{(1+r)^T}$$

Example 4:

What is the YTM of a 7.5% annual coupon bond, with a $1,000 face value, which matures in 3 years? The market price of the bond is currently $1,167.27.

YTM vs Coupon Rate

YTM includes the return from the coupon payments plus any capital gains on the bond (the difference between price and par value)

- Coupon rate is set by issuing company when bond is first sold. How?
- The relationship between the two helps categorize the bond

Type of Bond	Coupon Rate vs Yield to Maturity	Price Relationship to Par Value
Premium Bond	Coupon rate > Yield to Maturity	Price > Par Value
Par Value Bond	Coupon rate = Yield to Maturity	Price = Par Value
Discount Bond	Coupon rate < Yield to Maturity	Price < Par Value

Bond Price Over Time

As a bond matures, its price approaches face valuing since there are fewer coupon payments remaining.

Example 5:

A bond increases in price from $963.80 to $1,380.50 and pays a coupon of 10% (on a par value of $1,000) during the same period. What is the rate of return?

Callable Bonds

Bonds that can be redeemed or “paid back” by the issuer before the maturity date

- Issuer pays the call price for the bond, which is usually par/face value
- Issuer sometimes pays a call premium to the holder (some number of bonus coupon payments)
- Issuer will also specify a call window or call protection

Example 6:

Two years ago, the Mid-Atlantic Corporation issued a 10% coupon (paid semi-annually), 20-year maturity, bond with a 5-year deferred call window and a call premium of one coupon payment in addition to the par value ($1000) if exercised. If the current price on these bonds is $1080, what is its yield to call?

Corporate Bonds and Risk

Most U.S. federal bonds are considered safe and have no risk of default.

- **Default (credit) risk** involves the bond issuer not paying on its bond obligations (either coupon payments or face value)
- The **default premium** is the difference between the return on federal bonds (safe) and corporate bonds (risky).
- **Credit rating agencies** (like Moody's and Standards and Poor's) provide insight into a bond's risk with their rating scale.

Bond Ratings

Common ratings from two major credit rating firms. Note the terminology associated with the groupings of ratings.

Moody's	Standard & Poor's	Safety	Category
Aaa	AAA	The strongest rating; ability to repay interest and principal is very strong.	Investment Grade
Aa	AA	Very strong likelihood that interest and principal will be repaid.	
A	A	Strong ability to repay, but some vulnerability to changes in circumstances.	
Baa	BBB	Adequate capacity to repay; more vulnerability to changes in economic circumstances.	
Ba	BB	Considerable uncertainty about ability to repay.	Junk Bonds
B	B	Likelihood of interest and principal payments over sustained periods is questionable.	
Caa Ca	CCC CC	Bonds that may already be in default or in danger of imminent default.	
C	C	Little prospect for interest or principal on the debt ever to be repaid.	

Dealing with Bond Risk

The bond market has developed tools to manage the risk associated with corporate bonds.

- **Seniority:** a ranking of debt priority to be serviced. A senior claim is paid first before any subordinate (junior) claims
- **Secured debt** has a claim on specific assets (collateral) of the firm
- **Protective covenants** are restriction, imposed by the bondholders on the issuer, on the use of borrowed funds